CSC5035Z NLP 2026 — Assignment 1: Text Classification

Intent Detection in African Languages

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Reproducibility note: All reported metrics, tables and figures were generated from files in the outputs/ directory produced by the submitted code. Running `main.py` with the provided configuration reproduces the reported results (subject to standard floating-point/runtime variation).

1 Language Background

The selected languages for the project are **Amharic** (`amh`) and **Zulu** (`zul`), chosen to maximise typological contrast. Amharic is a Semitic language spoken by people in Ethiopia. It uses the **Ethiopic (Ge'ez) script**, where each character typically represent a consonant-vowel unit. This can provide compact and informative subword structure for NLP tokenizations. Zulu is a **Bantu language** spoken in South Africa and written in Latin script. It is strongly agglutinative, meaning words are built by attaching multiple prefixes and suffixes to a root. This structural differences are expected to produce different behavior across tokenization and classification pipelines. I expected Amharic to be more stable because of recurring word chunks can be strongly associated with intents, while Zulu would benefit more from subword and character-based representations.

2 Results Summary

Table 1: Dev and test accuracy (%) for all models on both languages. Best hyperparameter values are also shown.

	Hyperparameters		Amharic (amh)		Zulu (zul)	
Model	**amh**	**zul**	Dev	Test	Dev	Test
LR – Word unigrams	C=5.0	C=10.0	88.1	86.9	76.6	79.2
LR – Char n-grams	n=$(1,3)$, C=0.1	n=$(3,5)$, C=5.0	93.4	90.5	**81.3**	**85.3**
NB – BPE	k=100, α=1.0	k=750, α=1.0	90.9	86.6	78.1	82.0
LR – BPE only	k=500, C=0.5	k=750, C=0.5	92.5	90.3	78.1	81.3
LR – BPE + Bigram	k=100, C=0.5	k=100, C=10.0	**94.1**	91.7	**81.3**	84.7
LR – BPE + Length	k=500, C=0.5	k=500, C=1.0	92.5	90.3	77.8	82.0
LR – BPE + Bigram + Length	k=100, C=0.5	k=100, C=0.5	**94.1**	**91.9**	80.9	84.8

Overall pattern: Performance differs by language and feature type. For Amharic, the best model is LR with BPE based engineered features (94.1% dev, 91.9% test). For Zulu, the best model is LR with character n-grams (81.3% dev, 85.3% test), with BPE variants only marginally behind.

NB vs LR: LR consistently outperforms NB across both languages. A likely explanation is that LR can exploit feature interactions more effectively, whereas multinomial NB assumes conditional independence between tokens given the class. NB remains competitive on the development set, particularly for Amharic, but generalises less well to the test set, likely because overlapping intent vocabularies violate the independence assumption more severely at test time.

Language-dependent differences: The best hyperparameters differ markedly between languages. For example, NB performs best at k=100 for Amharic but k=750 for Zulu. This reinforces that optimal tokenization granularity is language-specific. This likely reflects the larger inventory of Ethiopic characters in the Amharic, where fewer merges are needed to form meaningful units. Overall, results support the view that model and feature choices should be adapted to language structure rather than assumed to transfer uniformly.

3 Tokenizer Analysis

Key observations from tokenization:

At k=100, both languages show a heavy fragmentation, but the pattern differs.Amharic utterances are split almost entirely into individual Ethiopic characters, reflecting the large inventory of the Ge'ez script. Zulu, using the Latin script, starts from a smaller alphabet and already shows early sub-word merges such as *th* and *cela* at k=100, indicating faster initial consolidation of common verb stems and prefixes, consistent with the agglutinative structure.

As k increases, both languages show less granular segmentation, though the pattern is not uniform. At k=300, Amharic shows stronger consolidation of common words.**ማንቂያ** (alarm) and **ሰአት** (hour) already appear as single tokens, while Zulu still fragments many stems. This is notable because *i-alamu* (alarm) appears in 4 out of 5 Zulu examples yet does not consolidate until k=500, whereas **ማንቂያ** merges by k=300 despite similar or even less frequency. A likely explanation is that Ethiopic characters are more complex units than Latin characters, so BPE needs fewer total merges to reconstruct a whole Amharic word from its characters than it does for a Latin-script word of similar length.

Table 2: BPE tokenization examples for Amharic (amh). Truncated for space.

Utterance	k=100	k=300	k=500
ለ9 ሰአት አንድ እና	ለ, 9, _, ሰአት_, አ, ን, ...	ለ, 9, _, ሰአት_, አ, ንድ_, ...	ለ, 9, _, ሰአት_, አንድ_, እና_, ...
ማንቂያ ደውል መቅጠር	ማ, ን, ቂ, ያ_, ደ, ው, ል_, ...	ማንቂያ_, ደው, ል_, መ, ቅ, ጠ, ር_, ...	ማንቂያ_, ደውል_, መ, ቅጠ, ር_, እፈልጋ, ለው_
ማንቂያ መቅጠር	ማ, ን, ቂ, ያ_, መ, ቅ, ጠ, ር_, ...	ማንቂያ_, መ, ቅ, ጠ, ር_, ፈል, ጌ, _, ...	ማንቂያ_, መ, ቅጠ, ር_, ፈል, ጌ_, ነበ, ር_
ነገ 9 ሰአት እንድወጣ	ነ, ገ, _, 9, _, ሰአት_, እን, ድ, ...	ነገ, _, 9, _, ሰአት_, እንድ, ወጣ, _, ...	ነገ_, 9, _, ሰአት_, እንድ, ወጣ, _, ማንቂያ_, ...
እባክህ ለጠዋት 12	እባክ, ህ_, ለ, ጠ, ዋ, ት_, 1, 2, _, ...	እባክህ_, ለ, ጠ, ዋ, ት_, 1, 2, _, ...	እባክህ_, ለ, ጠዋት_, 1, 2_, ሰአት_, ማንቂያ_, ...

Table 3: BPE tokenization examples for Zulu (zul). Truncated for space.

Utterance	k=100	k=300	k=500
Setha i-alamu ngo-5:30	S, e, th, a_, i-, al, am, ...	S, eth, a_, i-, al, am, ...	S, etha_, i-alamu_, ngo-, ...
Sicela usethe i-alamu	S, i, cela_, us, e, th, e_, ...	S, i, cela_, us, eth, e_, ...	S, i, cela_, us, eth, e_, ...
Setha i-alamu ngo-7	S, e, th, a_, i-, al, am, ...	S, eth, a_, i-, al, am, ...	S, etha_, i-alamu_, ngo-, ...
Ngivuse ngo-6 ekuseni	Ngi, v, us, e_, ngo, -, 6, ...	Ngi, v, us, e_, ngo-, 6, ...	Ngi, v, use_, ngo-, 6, ...
Setha i-alamu yango-6	S, e, th, a_, i-, al, am, ...	S, eth, a_, i-, al, am, ...	S, etha_, i-alamu_, y, ang, ...

4 Hyperparameter Analysis

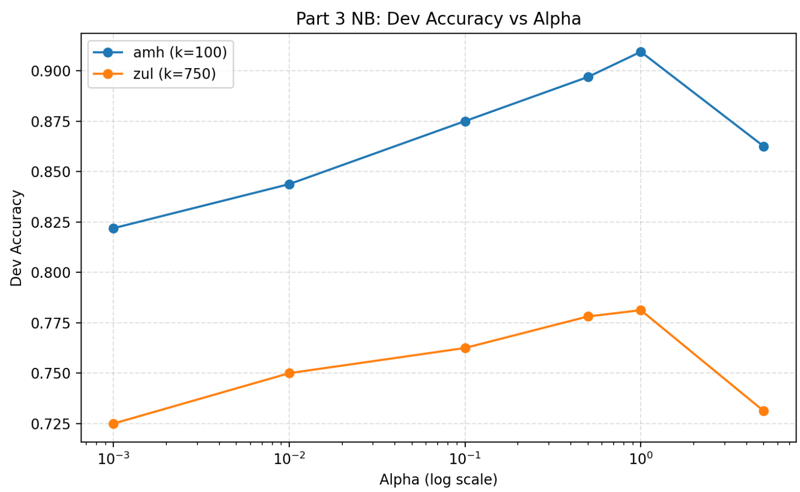

Figure 1: NB Dev accuracy as a function of Laplace smoothing α for both languages.

The dev-accuracy curves show the same qualitative shape in both languages: performance rises from very small α to a peak at α=1, then declines for larger α.

A very small value means weak smoothing. Token likelihoods are dominated by raw counts, and rare tokens get unstable probabilities, which may cause overfitting.

A large value over-smooths the distributions and class-conditional token probabilities can become too similar, reducing the representation power and causing underfitting.

Although the shape is consistent, the gain from tuning α is larger for Amharic than for Zulu (+7.5 points vs. +5 points). This may suggest that Amharic NB performance is more sensitive to smoothing strength under the selected tokenization setting, while Zulu remains comparatively harder for NB to optimize. **Note:** k is fixed, set to when it achieved the best dev accuracy for each language.

5 Error Analysis

Table 4: Top-3 confused intent pairs for the best model on each language (test set).

Lang	True intent → Predicted intent	Count	Model
amh	`recipe` → `meal_suggestion`	5	LR - BPE + Bigram + Length
amh	`play_music` → `update_playlist`	4	LR - BPE + Bigram + Length
amh	`recipe` → `ingredients_list`	3	LR - BPE + Bigram + Length
zul	`spending_history` → `transactions`	8	LR - Char n-gram
zul	`min_payment` → `bill_balance`	4	LR - Char n-gram
zul	`weather` → `book_hotel`	4	LR - Char n-gram

Table 5: Misclassified examples for top confused intent pairs (test set).

True	Predicted	Utterance
Amharic		
`recipe`	`meal_sug.`	አጥሚት እንዴት መስራት እንዴት እንዳለብኝ አስረዳኝ□
`recipe`	`meal_sug.`	ቦርዴ ከምን ምግብ ጋር አብሮ ይሄዳል□
`play_music`	`upd_playlist`	በጣም የፈለኩት ነገር ቢኖር የአለማየሁ እሸቴን ሙዚቃዎች መስማት ነው።
`play_music`	`upd_playlist`	የቀዳማዊት እመቤት ዝናሽ ታያቸውን መዝሙር ዝርዝር አስጀምረው።
`recipe`	`ingr_list`	ቅቤን ለማንጠር የሚያስፈልጉን ቅመሞች ምን ምን ናቸው□
`recipe`	`ingr_list`	ቆንጆ ሽሮ ለመስራት መሰረታዊ ቅመሞች ምን ምን ናቸው□
Zulu		
`spend_hist.`	`transactions`	ngiyithenge ngamalini i-iPhone e-iStore izolo?
`spend_hist.`	`transactions`	yini ebiza kakhulu engiyikhokhelile kulenyanga phakathi kokudla nezingubo zokugqoka?
`min_payment`	`bill_balance`	imalini ekumele ifakwe kwibhili yami yakaTru-worths?
`min_payment`	`bill_balance`	kumele ngikhokhe malini kwi-account yami yak-waVirgin Active?
`weather`	`book_hotel`	Lingakanani izinga lokushisa eLuanda njenga-manje?
`weather`	`book_hotel`	Sinjani isimo sezulu eGoli ntambama?

Token distribution analysis: For each confused intent pair, I extracted the top 10 positively weighted features for each intent from the trained classifier and measured overlap.

This analysis yielded limited meaningful insights for the best-performing models. In Amharic errors concentrate in semantically adjacent intent pairs (primarily food/music). With `LR with all features` and k = 100 as the best model/feature/hyperparam set, many top-ranked features are short subword fragments, which are difficult to interpret semantically; observed overlap therefore reflects shared subword structure more than clear intent-level linguistic similarity. A similar limitation appears in Zulu. In Zulu, the best model `char n-grams (3,5)`, shows same general issue: top features are often short character fragments, and confusion appears to be driven more by semantic closeness between intent labels than by clearly interpretable token-distribution overlap.

1.**[amh]** `recipe/meal_suggestion`: There is limited but not negligible overlap (e.g.,*bpe*=የሚ). This indicates partial lexical sharing, but not feature collapse, so confusion likely comes from requests that mix for instance "which ingredient" and "what to eat" semantic.

2.**[amh]** `play_music/update_playlist`: The pair has three overlapping top-ranked BPE features, but mostly short subword fragments. The ovserved misclassifications are more plausibly explained utterance-level ambiguity.

3.**[amh]** `ingredient_list/recipe`: No overlapping features, but both intents contain ingredient and preparation vocabulary. Errors persist because of thin semantic boundary.

1.**[zul]** `spending_history/transactions`: No overlap between n-gram features. Likely features that can appear in both are words that translates to "money" and "pay".

2.**[zul]** `min_payment/bill_balance`: No overlap between n-gram features. Likely features that can appear in both are words that translates to ”money” and ”pay”.

3.**[zul]** `book_hotel/weather`: No overlap between n-gram features. Could be because both intents often contain place names and location words.

The clearest cross-language difference is domain-specific: Amharic errors cluster in the food/meal intents, while Zulu errors cluster in transaction/money intents. This suggests stronger semantic adjacency within those domain labels in each language's data, rather than purely language-structural effect.

6 Feature Engineering

Feature 1: BPE Token Bigrams

I added adjacent BPE-token bigram features on top of BPE counts to capture short local context that unigram counts miss. This was motivated by repeated multi-token expressions in the training data, where intent cues often appear as short sequences rather than isolated tokens. Empirically, this feature gave the strongest improvement and produced the best development performance for both languages, showing that local token order carries useful intent information.

Feature 2: Length/Structure Features

I added three scalar features per utterance: number of whitespace tokens, number of characters, and average token length. The motivation was that some intents appeared to differ in utterance shape (short commands vs longer commands). In practice, this feature set gave little to no gain over BPE counts alone. A likely reason is that these three global features are weak compared to high-dimensional lexical/subword features, so their contribution becomes small in the final classifier.

Overall, BPE bigrams were useful and retained, while length/structure features had negligible standalone value in this setup.

AI Tool Usage

AI assistance was used for debugging support, project scaffold/structure and occasional clarification of reporting questions. All architecture choices, experiments, analysis and final submitted code were completed and verified by me.